XRPL Validator — Daily Engineering Report · 2026-08-30

Period: 2026-08-30 00:00 to 24:00 US Eastern (2026-08-30 04:00 to 2026-08-31 04:00 UTC); ledgers #106,636,602 to #106,658,957. Every count from the ledger and the validator's logs stops at #106,658,957. Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: the day's first and last ledger, from a public full-history server; Step 1 receipt records (they begin 2026-08-29); xrpl-validator merge records (they begin 2026-07-04). Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); amendment data; data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** 3 engine receipts recorded (2 fixed, 0 explained, 1 without a recorded fix).
- **Changes:** 0 deployments, 5 commits merged into main.

Service levels

No signing records for this day (they begin 2026-09-19).

Native engine conformance

Ledgers with a receipt in the period (#106,636,602–#106,658,957), by step:

Step	Ledgers with a receipt
Step 1 · native engine	3
Step 2 · records	0
Step 3 · order	0

Step	Ledgers with a receipt
Step 4 · peer data	0
Step 4 · validations	0
Step 5 · own hash	0
Writer	0

Live counters are kept only for the current run; none exist for a past day.

Receipt register

ID	Time	Ledger(s)	Step	Description	Disposition
R-01	—	#106,648,821	Step 1 · native engine	150 ledger change(s) the network made were absent from ours: F4173CAD, 5275D0F2, BFAAA2E3	no disposition recorded
R-02	—	#106,648,822	Step 1 · native engine	119 ledger change(s) the network made were absent from ours: 1363CBD9, E1334AEB, AE7618F9	fixed: commit 6a853fd, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-03	—	#106,648,823	Step 1 · native engine	100 ledger change(s) the network made were absent from ours: ABB8ABA5, 528FAE4E, 13969DF2; 1 change(s) of ours the network did not make	fixed: commit 6a853fd, deployed in cycle 121 (2026-09-11 17:39 UTC)

A receipt is recorded whenever the validator's own work differs from the network's, or the network's trusted validations reached it short of quorum. One ledger flagged by several steps is one receipt. Dispositions: fixed (with the finding, commit and deployment), explained (not a defect, with the reason), or open. Receipt and incident IDs are numbered per report: R-01 here is R-2026-08-30-01, INC-01 is INC-2026-08-30-01. Times are not shown where the reference node no longer holds the ledger.

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

Merged	Repo	Commit	Subject
13:35 UTC	xrpl-validator	`99bd44a`	fix(xrpl-ledger): XRP delivery re-arms the destination's free key reset
14:54 UTC	xrpl-validator	`ba4ae17`	fix(xrpl-ledger): LP deposit sum rounds half-even at 16 digits
16:52 UTC	xrpl-validator	`1af1d6d`	feat(scripts): durable VHS listing watch (systemd user timer, 30min)
16:52 UTC	xrpl-validator	`82df826`	fix(ledger): regross the want pot from the net remainder each iteration (finding 37)
17:01 UTC	xrpl-validator	`2bb284f`	feat(node): shadow hydration v5 memory diet (budget gate, no-cache scan, RSS receipts)

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **fix(xrpl-ledger): XRP delivery re-arms the destination's free key reset** (`99bd44a`, xrpl-validator) Finding 35 (#106644275, dst rMWEptMt): receiving an XRP payment clears the destination's lsfPasswordSpent — 'Re-arm the password change fee if we can and need to', Payment.cpp:674-676, pinned by SetRegularKey_test.cpp:98-100 (fee-0 regkey sets the flag, a later inbound payment clears it). Ours never cleared, and PasswordSpent accounts receive XRP constantly: the fresh window cascaded from ledger two. 2/61 -> 46/60; …
- **fix(xrpl-ledger): LP deposit sum rounds half-even at 16 digits** (`ba4ae17`, xrpl-validator) Finding 36 (#106644320 749D3E45, tfTwoAsset): the minted LP amount was already byte-exact — the divergence was THE ADD. The exact sum 279228264.81815046 must 16-digit-round nearest (truth …1505); both the old exact-string path and a downward round land …1504. Receipts (new DX_AMMDEP: minted/lp_pre/lp_post/sized arm) pinned it after two model-driven theories missed: the revival-initializer port (kept — correct per …
- **feat(scripts): durable VHS listing watch (systemd user timer, 30min)** (`1af1d6d`, xrpl-validator) We are absent from the VHS validator list (manifest rejected: master == ephemeral). Session-bound watches die silently, so a cron-style timer polls data.xrpl.org and telegrams once per state change: first listing, server_version rendering, or de-listing. Timer offset :07/:37 keeps clear of the 03:00 batch window.
- **fix(ledger): regross the want pot from the net remainder each iteration (finding 37)** (`82df826`, xrpl-validator) rippled's remainingOut lives NET (savedOuts in outReq units) and every flow iteration regrosses it through the last DirectStep's rev pass with mulRatio roundUp. Our gross-subtraction pot (want_gross minus grossed fills) cancels tails the fresh product keeps: #106644326 431CAAFC iter-5 net 1.442557458781219 x 1.001 = ...240000|219 regrosses to ...240001 while the pot said ...240000, and the UNI/XRPS book rev-sizing …
- **feat(node): shadow hydration v5 memory diet (budget gate, no-cache scan, RSS receipts)** (`2bb284f`, xrpl-validator) v4's single greedy pass spiked live_viewer to 58.6GB on the 62GB box and the machine thrashed; the mirror itself (~13-15GB) fits — the SPIKE is the killer. Three levers: - Budget gate: hydrate refuses to start unless /proc/meminfo MemAvailable >= XRPL_SHADOW_HYDRATE_MIN_GB (default 20). A skipped hydrate is a receipt (hydrate_skipped_lowmem stat + log line) and re-arms the 600s cooldown; a thrashing validator is an …

Independence program

No independence record for UTC day 2026-08-30; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

- Receipt R-01 (#106,648,821): open, under investigation.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Cooling-Off Custody Withdrawal Holds** (custody) Add an AccountPolicy ledger object (ltACCOUNT_POLICY, owner-reserved, keylet::accountPolicy(account)) set via a new AccountPolicySet transaction carry
- **Deterministic Jittered Iceberg Replenishment** (dex) Extend g31's tfIceberg OfferCreate flag by replacing the fixed refill size with sfMinReplenish and sfMaxReplenish SFields alongside sfDisplayQuantity.
- **Bounded WASM Hooks for Mainnet** (hooks) Port Xahau's proven Hooks execution model to XRPL mainnet as the first concrete implementation of the in-flight XLS-101/XLS-102 line: add a Hook ledge

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
08-30 13:35	08-30 13:35	xrpl-validator	`99bd44a`	fix(xrpl-ledger): XRP delivery re-arms the destination's free key reset
08-30 14:54	08-30 14:54	xrpl-validator	`ba4ae17`	fix(xrpl-ledger): LP deposit sum rounds half-even at 16 digits
08-30 16:52	08-30 16:52	xrpl-validator	`1af1d6d`	feat(scripts): durable VHS listing watch (systemd user timer, 30min)
08-30 16:52	08-30 16:52	xrpl-validator	`82df826`	fix(ledger): regross the want pot from the net remainder each iteration (finding 37)
08-30 17:01	08-30 17:01	xrpl-validator	`2bb284f`	feat(node): shadow hydration v5 memory diet (budget gate, no-cache scan, RSS receipts)

Times in UTC. A commit appears on the day it was committed to main or merged into it.